

Lecture-2

HTML Notes: Structured Class Notes

Agenda

1. Setup for Web Development
 2. HTML Basics
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Setup for Web Dev

VSCode Download URL : <https://code.visualstudio.com/>

Essential Extensions:

- **Auto Rename Tag** – Automatically renames matching closing tags when you rename an opening tag
 - **Live Server** – Instantly previews your webpage changes in the browser
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What is HTML?

HTML = HyperText Markup Language

- Defines the content structure of a web page
 - File extension: `.html`
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HTML Tags

Basic Syntax:

It starts with a tag and in between you will have content and then closing tag

TagName : It gives behaviour to the tag

content : it is the content that is visible to the user

```
<tagstart>content</tagend>
```

Example:

```
<p>This is a paragraph</p>
```

Heading Tags

Headings create a hierarchical structure for content, from main titles to sub-sections.

```
<h1>I am h1</h1>
<h2>I am h2</h2>
<h3>I am h3</h3>
<h4>I am h4</h4>
<h5>I am h5</h5>
<h6>I am h6</h6>
```

Hierarchy Structure:

```
<h1> The Hound of Baskerville
  <h2> Chapter 1
    <h3> Subchapters
      <h4> Topic
        <h5> Detail
          <h6> Specific Info
```

Paragraph Tag

Used to group sentences and blocks of text.

```
<p>I am a paragraph</p>
```

Self-Closing Tags (Void Tags)

These tags do not wrap content. They perform a single action and close themselves.

Line Break `<br/>`

Inserts a line break (new line) in text.

```
<br/>
```

Horizontal Rule `<hr/>`

Draws a horizontal line across the page.

```
<hr/>
```

Code Example:

```
<p>I am p1</p>  
<br/>  
<br/>  
<p>I am p2</p>  
<hr/>
```

Lists

Lists organize items in either ordered (numbered) or unordered (bullet point) format.

Unordered List `<ul>`

Items are displayed with bullet points. Order does not matter.

```
<ul>  
  <li>Strawberry</li>  
  <li>Dragonfruit</li>  
  <li>Avocado</li>  
</ul>
```

Output:

- Strawberry
- Dragonfruit
- Avocado

Ordered List `<ol>`

Items are displayed with numbers. Order matters.

```
<ol>  
  <li>Iron Man</li>  
  <li>First Avenger</li>  
  <li>Thor</li>  
</ol>
```

Output:

1. Iron Man
2. First Avenger
3. Thor

List Items `<li>`

Represents a single item within an ordered or unordered list.

HTML Attributes

Attributes modify the default properties of HTML tags.

Syntax:

```
<tag key="value">content</tag>
```

Key Points:

- Attributes are placed in the **opening tag**
 - Format: `key="value"`
 - They modify the default behavior or properties of the tag
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Anchor Tag `<a>` (Hyperlink)

Used to create hyperlinks to other pages or locations.

Basic Syntax:

```
<a href="URL">Clickable Text</a>
```

Common Attributes:

- `href` – The destination URL (required)
- `target` – Where to open the link
 - `_blank` – Opens in a new tab
 - `_self` – Opens in the same tab

Code Example:

```
<!-- Opens Google in a NEW tab -->  
<a href="https://www.google.com/" target="_blank">
```

```
    Click to go somewhere
</a>

<!-- Opens Google in the SAME tab -->
<a href="https://www.google.com/" target="_self">
    Click to stay here
</a>
```

Quick Reference Table

Tag	Type	Purpose	Example
<h1> - <h6>	Heading	Create titles and section headings	Title
<p>	Paragraph	Group text content	Text
 	Self-closing	Insert line break	
	Self-closing	Draw horizontal line	
	Unordered list	Create bullet list	Item
	Ordered list	Create numbered list	Step 1
	List item	Single item in a list	Milk
<a>	Anchor	Create hyperlink	Click

Summary

1. HTML provides the structure for web pages
 2. Tags are used to define different types of content
 3. Headings (h1 to h6) show content hierarchy
 4. Lists organize items in ordered or unordered format
 5. Self-closing tags (br, hr) perform single actions
 6. Attributes customize tag behavior
 7. Anchor tags create hyperlinks to other pages
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